

29 April 2026

Exercise 2

01 ~~SELECT DISTINCT department~~

~~FROM students~~

The student table

01 SELECT DISTINCT department

FROM Student;

department
IT
HR

02 SELECT department

AVG (age) AS avg-age

FROM students

GROUP BY department

department	avg-age
IT	20
HR	22
Finance	23

03 SELECT department

COUNT (*) AS student

FROM students

GROUP BY department

HAVING COUNT (*) > 1

department	avg - age	student - count
IT	21,5	2
HR	21,5	2

04) SELECT student-id, name, age, department

FROM students

WHERE age BETWEEN 21 AND 23;

Student-id	Name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

05) SELECT student id, name, age, department

FROM student

WHERE department IN ('IT', 'HR')

AND age > 21

Student-id	name	age	department
4	Diana	23	IT
5	Eve	22	HR

Q2 THE COURSES TABLE

Q6) SELECT department
SUM(credits) AS total-credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;

department	total-credits
IT	11

Q7) SELECT course-id
FROM course
WHERE credits < 4;

course-id	course-name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Q8) SELECT course-id, course-name, department, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

course-id	course-name	department	credits
102	Python	IT	4
103	Data Science	IT	4
101	SQL Basics	IT	3

03 The enrollments Table

09) SELECT

MAX (grade) AS MAX - grade,
MIN (grade) AS MIN - grade,
AVG (grade) AS avg - grade,
FROM enrollments;

max - grade	min - grade	avg - grade
90	78	82

10) SELECT

course - id,
COUNT (*) AS enrollment - count
FROM enrollments
GROUP BY course - id;

course - id	enrollment - count
101	1
102	1
103	1
104	1
105	1

The enrollment - count
It will be (1) because
all the column appears
once like each group
has only 1 row

11) SELECT department,

SUM (salary) AS total - salary,
SUM (bonus) AS total - bonus

FROM salaries

(GROUP BY department

department	total - salary	total - bonus
IT	122000	11000
HR	109000	8500
Finance	70000	4000

12) SELECT

department

AVG (salary) AS avg-salary

FROM salaries

GROUP BY department

HAVING AVG (salary) > 5500

department	avg-salary
IT	61000
Finance	70000

13) SELECT

employee-id;

name;

salary,

Bonus

(salary + bonus) AS total-compensation

FROM salaries

WHERE (salary + bonus) > 60000

employee-id	name	salary	bonus	total-compensation
1	Tom	60000	5000	65000
3	Spike	70000	4000	74000
4	Tyke	62000	6000	68000

05) THE PROJECTS Table

14) SELECT department,
Sum (budget) AS total-budget
AVG (budget) AS Avg - Budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70000

department	total - budget	avg - budget
IT	270000	135000
finance	80000	80000

15) SELECT

project - id,
project - name,
department,
budget

FROM projects

WHERE budgets BETWEEN 80000 AND 120000

AND department <> 'marketing';

Project_id	Project-name	department	budget
1	AI App	IT	120000
2	Payrolls System	Finance	80000
5	HR Portal	HR	50000